

$$= 100000 + (60 - 40) = 100020.$$

$$\therefore N = (100020 + 2) = 100022.$$

$$\text{Sum of digits in } N = (1 + 2 + 2) = 5.$$

- 104.** L.C.M. of 4, 5, 6, 7 and 8 is 840.

Let the required number be $840k + 2$, which is a multiple of 13.

Least value of k for which $(840k + 2)$ is divisible by 13 is $k = 3$.

$$\therefore \text{Required number} = 840 \times 3 + 2 = 2522.$$

- 105.** Here $(12 - 7) = 5$, $(15 - 10) = 5$ and $(16 - 11) = 5$.

$$\therefore \text{Required number} = (\text{L.C.M. of } 12, 15, 16) - 5 = 240 - 5 = 235.$$

- 106.** Here $(48 - 38) = 10$, $(60 - 50) = 10$, $(72 - 62) = 10$
 $(108 - 98) = 10$ & $(140 - 130) = 10$.

$$\therefore \text{Required number} = (\text{L.C.M. of } 48, 60, 72, 108, 140) - 10 = 15120 - 10 = 15110.$$

- 107.** Here $(18 - 7) = 11$, $(21 - 10) = 11$ and $(24 - 13) = 11$.
 L.C.M. of 18, 21 and 24 is 504.

Let the required number be $504k - 11$.

Least value of k for which $(504k - 11)$ is divisible by 23 is $k = 6$.

$$\therefore \text{Required number} = 504 \times 6 - 11 = 3024 - 11 = 3013.$$

- 108.** Clearly, $(2 - 1) = 1$, $(3 - 2) = 1$ and $(4 - 3) = 1$.

L.C.M. of 2, 3, 4 = 12.

So, the sequence shall have numbers of the form $12k - 1$, where $k = 1, 2, 3, \dots$

$$\therefore \text{Third term of the sequence} = 12 \times 3 - 1 = 36 - 1 = 35.$$

- 109.** Clearly, $(4 - 1) = 3$, $(5 - 2) = 3$, $(6 - 3) = 3$, $(7 - 4) = 3$ and $(8 - 5) = 3$.

L.C.M. of 4, 5, 6, 7, 8 = 840.

Greatest number of 4 digits = 9999.

On dividing 9999 by 840, the remainder is 759.

So, required number = $(9999 - 759) - 3 = 9237$.

- 110.** L.C.M. of 5, 6, 7, 8 = 840.

$\therefore$ Required number is of the form $840k + 3$.

Least value of k for which $(840k + 3)$ is divisible by 9 is $k = 2$.

$$\therefore \text{Required number} = (840 \times 2 + 3) = 1683.$$

- 111.** L.C.M. of 16, 18, 20, 25 = 3600.

Required number is of the form $3600k + 4$.

Least value of k for which $(3600k + 4)$ is divisible by 7 is $k = 5$.

$$\therefore \text{Required number} = (3600 \times 5 + 4) = 18004.$$

- 112.** L.C.M. of 6, 8, 10, 12 = 120.

$\therefore$ Required number is of the form $120k + 5$.

Least value of k for which $(120k + 5)$ is divisible by 13 is $k = 7$.

$$\therefore \text{Required number} = (120 \times 7 + 5) = 845.$$

- 113.** Required number of beads = L.C.M. of 16, 20, 36 = 720.

- 114.** Interval of change = (L.C.M. of 60 and 62) sec = 1860 sec = 31 min.

So, the devices would beep together 31 min after 10 a.m., i.e., at 10.31 a.m.

- 115.** L.C.M. of 2, 4, 6, 8, 10, 12 is 120.

So, the bells will toll together after every 120 seconds, i.e., 2 minutes.

$$\text{In 30 minutes, they will toll together } \left[\left(\frac{30}{2} \right) + 1 \right] = 16 \text{ times.}$$

- 116.** Interval after which the bells will toll together

$$= (\text{L.C.M. of } 6, 7, 8, 9) \text{ sec} = 504 \text{ sec.}$$

$$\text{In 1.54 hours, they will toll together } \left[\left(\frac{1.54 \times 60 \times 60}{504} \right) + 1 \right]$$

times = 12 times.

- 117.** Interval after which the devices will beep together

$$= (\text{L.C.M. of } 30, 60, 90, 105) \text{ min.}$$

$$= 1260 \text{ min} = 21 \text{ hrs.}$$

So, the devices will again beep together 21 hrs. after 12 noon i.e., at 9 a.m.

- 118.** L.C.M. of 24, 36, 48 = 144.

So, the three girls will meet at one point in 144 seconds i.e., 2 min 24 sec.

- 119.** L.C.M. of $\frac{91}{6}$, $\frac{65}{4}$ and $\frac{56}{3} = \frac{\text{L.C.M. of } 91, 65, 56}{\text{H.C.F. of } 6, 4, 3} = 3640$.

So, the three persons will be together at the starting point in 3640 sec

i.e., 1 hour 40 seconds.

- 120.** L.C.M. of 252, 308 and 198 = 2772.

So, A, B and C will again meet at the starting point in 2772 sec. i.e., 46 min. 12 sec.

- 121.** For one complete revolution, the first, second and third wheels take $\frac{60}{40}$, $\frac{60}{24}$, $\frac{60}{16}$ seconds i.e., $\frac{3}{2}$, $\frac{5}{2}$, $\frac{15}{4}$ seconds respectively.

$\therefore$ Time taken for all red spots to touch the ground again simultaneously.

$$= \left(\text{L.C.M. of } \frac{3}{2}, \frac{5}{2}, \frac{15}{4} \right) \text{ sec} = \left(\frac{\text{L.C.M. of } 3, 5, 15}{\text{H.C.F. of } 2, 2, 4} \right) \text{ sec} = \frac{15}{2} \text{ sec} = 7\frac{1}{2} \text{ sec.}$$

- 122.** First pendulum strikes once in $\frac{3}{5}$ seconds. Second pendulum strikes once in $\frac{4}{7}$ seconds.

$$\text{L.C.M. of } \frac{3}{5} \text{ and } \frac{4}{7} = \frac{\text{L.C.M. of } 3 \text{ and } 4}{\text{H.C.F. of } 5 \text{ and } 7} = 12.$$

So, they strike together after every 12 seconds.

$$\text{Thus, they strike together } \left(\frac{60}{12} + 1 \right) = 6 \text{ times in 1 minute.}$$

$\therefore$ Total number of clear strikes heard

$$= \left[\frac{60}{\left(\frac{3}{5} \right)} + \frac{60}{\left(\frac{4}{7} \right)} \right] - 6 = \left(60 \times \frac{5}{3} + 60 \times \frac{7}{4} \right) - 6 = (100 + 105) - 6 = 199.$$